

2.12 Exercises

Exercise 2.1: Discontinuous MPC

In Example 2.8, compute $\mathcal{U}_3(x)$, $V_3^0(x)$, and $\kappa_3(x)$ at a few points on the unit circle.

Exercise 2.2: Boundedness of discrete time model

Complete the proof of Proposition 2.16 by showing that $f(\cdot)$ and $f_{\mathbb{Z}}^{-1}(\cdot)$ are bounded on bounded sets.

Exercise 2.3: Destabilization with state constraints

Consider a state feedback regulation problem with the origin as the setpoint (Muske and Rawlings, 1993). Let the system be

$$A = \begin{bmatrix} 4/3 & -2/3 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad C = [-2/3 \ 1]$$

and the controller objective function tuning matrices be

$$Q = I \quad R = I \quad N = 5$$

- Plot the unconstrained regulator performance starting from initial condition $x(0) = \begin{bmatrix} 3 & 3 \end{bmatrix}'$.
- Add the output constraint $y(k) \leq 0.5$. Plot the response of the constrained regulator (both input and output). Is this regulator stabilizing? Can you modify the tuning parameters Q, R to affect stability as in Section 1.3.4?
- Change the output constraint to $y(k) \leq 1 + \epsilon, \epsilon > 0$. Plot the closed-loop response for a variety of ϵ . Are any of these regulators destabilizing?
- Set the output constraint back to $y(k) \leq 0.5$ and add the terminal constraint $x(N) = 0$. What is the solution to the regulator problem in this case? Increase the horizon N . Does this problem eventually go away?

Exercise 2.4: Computing the projection of $\mathbb{Z}$ onto $\mathcal{X}_N$

Given a polytope

$$\mathbb{Z} := \{(x, u) \in \mathbb{R}^n \times \mathbb{R}^m \mid Gx + Hu \leq \psi\}$$

write an Octave or MATLAB program to determine $\mathcal{X}$, the projection of $\mathbb{Z}$ onto $\mathbb{R}^n$

$$\mathcal{X} = \{x \in \mathbb{R}^n \mid \exists u \in \mathbb{R}^m \text{ such that } (x, u) \in \mathbb{Z}\}$$

Use algorithms 3.1 and 3.2 in Keerthi and Gilbert (1987).

To check your program, consider a system

$$x^+ = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

subject to the constraints $\mathbb{X} = \{x \mid x_1 \leq 2\}$ and $\mathbb{U} = \{u \mid -1 \leq u \leq 1\}$. Consider the MPC problem with $N = 2$, $\mathbf{u} = (u(0), u(1))$, and the set $\mathbb{Z}$ given by

$$\mathbb{Z} = \{(x, \mathbf{u}) \mid x, \phi(1; x, \mathbf{u}), \phi(2; x, \mathbf{u}) \in \mathbb{X} \text{ and } u(0), u(1) \in \mathbb{U}\}$$

Verify that the set

$$\mathcal{X}_2 := \{x \in \mathbb{R}^2 \mid \exists u \in \mathbb{R}^2 \text{ such that } (x, u) \in \mathbb{Z}\}$$

is given by

$$\mathcal{X}_2 = \{x \in \mathbb{R}^2 \mid Px \leq p\} \quad P = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix} \quad p = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$$

Exercise 2.5: Computing the maximal output admissible set

Write an Octave or MATLAB program to determine the maximal constraint admissible set for the system $x^+ = Fx$, $y = Hx$ subject to the hard constraint $y \in Y$ in which $Y = \{y \mid Ey \leq e\}$. Use algorithm 3.2 in Gilbert and Tan (1991).

To check your program, verify for the system

$$F = \begin{bmatrix} 0.9 & 1 \\ 0 & 0.09 \end{bmatrix} \quad H = \begin{bmatrix} 1 & 1 \end{bmatrix}$$

subject to the constraint $Y = \{y \mid -1 \leq y \leq 1\}$, and that the maximal output admissible set is given by

$$O_\infty = \{x \in \mathbb{R}^2 \mid Ax \leq b\} \quad A = \begin{bmatrix} 1 & 1 \\ -1 & -1 \\ 0.9 & 1.09 \\ -0.9 & -1.09 \\ 0.81 & 0.9981 \\ -0.81 & -0.9981 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Show that t^* , the smallest integer t such that $O_t = O_\infty$ satisfies $t^* = 2$.

What happens to t^* as F_{22} increases and approaches one? What do you conclude for the case $F_{22} \geq 1$?

Exercise 2.6: Terminal constraint and region of attraction

Consider the system

$$x^+ = Ax + Bu$$

subject to the constraints

$$x \in \mathbb{X} \quad u \in \mathbb{U}$$

in which

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \mathbb{X} = \{x \in \mathbb{R}^2 \mid x_1 \leq 5\} \quad \mathbb{U} = \{u \in \mathbb{R}^2 \mid -1 \leq u \leq 1\}$$

and $\mathbf{1} \in \mathbb{R}^2$ is a vector of ones. The MPC cost function is

$$V_N(x, u) = \sum_{i=0}^{N-1} \ell(x(i), u(i)) + V_f(x(N))$$

in which

$$\ell(x, u) = (1/2)(|x|_Q^2 + |u|^2) \quad Q = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix}$$

and $V_f(\cdot)$ is the terminal penalty on the final state.

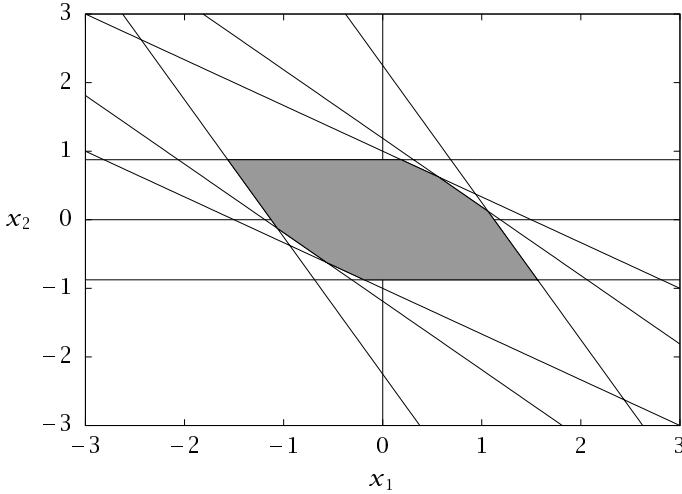


Figure 2.10: Region of attraction (shaded region) for constrained MPC controller of Exercise 2.6.

- (a) Implement unconstrained MPC with no terminal cost ($V_f(\cdot) = 0$) for a few values of α . Choose a value of α for which the resultant closed loop is unstable. Try $N = 3$.
- (b) Implement constrained MPC with no terminal cost or terminal constraint for the value of α obtained in the previous part. Is the resultant closed loop stable or unstable?
- (c) Implement constrained MPC with terminal equality constraint $x(N) = 0$ for the same value of α . Find the region of attraction for the constrained MPC controller using the projection algorithm from Exercise 2.4. The result should resemble Figure 2.10.

Exercise 2.7: Infinite horizon cost to go as terminal penalty

Consider the system

$$x^+ = Ax + Bu$$

subject to the constraints

$$x \in \mathbb{X} \quad u \in \mathbb{U}$$

in which

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$\mathbb{X} = \{x \in \mathbb{R}^2 \mid -5 \leq x_1 \leq 5\} \quad \mathbb{U} = \{u \in \mathbb{R}^2 \mid -1 \leq u \leq 1\}$$

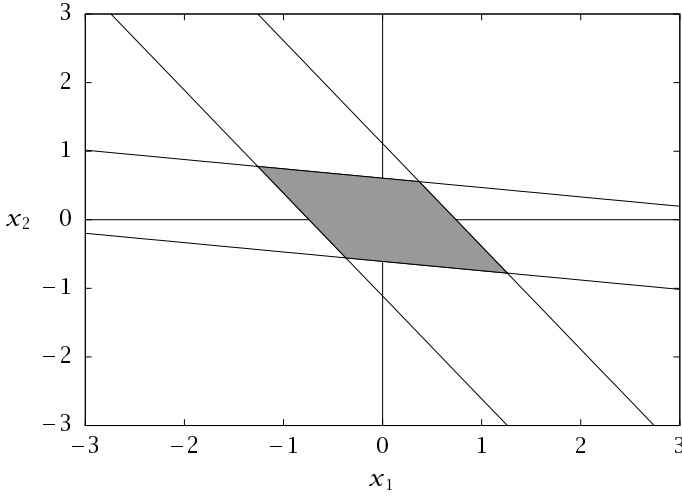


Figure 2.11: The region $\mathbb{X}_f$, in which the unconstrained LQR control law is feasible for Exercise 2.7.

The cost is

$$V_N(\mathbf{x}, \mathbf{u}) := \sum_{i=0}^{N-1} \ell(\mathbf{x}(i), \mathbf{u}(i)) + V_f(\mathbf{x}(N))$$

in which

$$\ell(\mathbf{x}, \mathbf{u}) = (1/2)(|\mathbf{x}|_Q^2 + |\mathbf{u}|^2) \quad Q = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix}$$

and $V_f(\cdot)$ is the terminal penalty on the final state and $\mathbf{1} \in \mathbb{R}^2$ is a vector of all ones. Use $\alpha = 10^{-5}$ and $N = 3$ and terminal cost $V_f(\mathbf{x}) = (1/2)\mathbf{x}'\Pi\mathbf{x}$ where Π is the solution to the steady-state Riccati equation.

- Compute the infinite horizon optimal cost and control law for the unconstrained system.
- Find the region $\mathbb{X}_f$, the maximal constraint admissible set using the algorithm in Exercise 2.5 for the system $\mathbf{x}^+ = (A + BK)\mathbf{x}$ with constraints $\mathbf{x} \in \mathbb{X}$ and $K\mathbf{x} \in \mathbb{U}$. You should obtain the region shown in Figure 2.11.
- Add a terminal constraint $\mathbf{x}(N) \in \mathbb{X}_f$ and implement constrained MPC. Find $\mathcal{X}_N$, the region of attraction for the MPC problem with $V_f(\cdot)$ as the terminal cost and $\mathbf{x}(N) \in \mathbb{X}_f$ as the terminal constraint. Contrast it with the region of attraction for the MPC problem in Exercise 2.6 with a terminal constraint $\mathbf{x}(N) = 0$.
- Estimate $\tilde{\mathcal{X}}_N$, the set of initial states for which the MPC control sequence for horizon N is equal to the MPC control sequence for an infinite horizon. Hint: $\mathbf{x} \in \tilde{\mathcal{X}}_N$ if $\mathbf{x}^0(N; \mathbf{x}) \in \text{int}(\mathbb{X}_f)$. Why?

Exercise 2.8: Terminal penalty with and without terminal constraint

Consider the system

$$x^+ = Ax + Bu$$

subject to the constraints

$$x \in \mathbb{X} \quad u \in \mathbb{U}$$

in which

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and

$$\mathbb{X} = \{x \in \mathbb{R}^2 \mid -15 \leq x_1 \leq 15\} \quad \mathbb{U} = \{u \in \mathbb{R}^2 \mid -5 \cdot \mathbf{1} \leq u \leq 5 \cdot \mathbf{1}\}$$

The cost is

$$V_N(x, \mathbf{u}) = \sum_{i=0}^{N-1} \ell(x(i), u(i)) + V_f(x(N))$$

in which

$$\ell(x, u) = (1/2)(|x|_Q^2 + |u|)^2 \quad Q = \begin{bmatrix} \alpha & 0 \\ 0 & \alpha \end{bmatrix}$$

$V_f(\cdot)$ is the terminal penalty on the final state, and $\mathbf{1} \in \mathbb{R}^2$ is a vector of ones.

Use $\alpha = 10^{-5}$ and $N = 3$ and terminal cost $V_f(x) = (1/2)x'Px$ where $V_f(\cdot)$ is the infinite horizon optimal cost for the unconstrained problem.

- Add a terminal constraint $x(N) \in \mathbb{X}_f$, in which $\mathbb{X}_f$ is the maximal constraint admissible set for the system $x^+ = (A + BK)x$ and K is the optimal controller gain for the unconstrained problem. Using the code developed in Exercise 2.7, estimate X_N , the region of attraction for the MPC problem with this terminal constraint and terminal cost. Also estimate $\hat{X}_N$, the region for which the MPC control sequence for horizon N is equal to the MPC control sequence for infinite horizon. Your results should resemble Figure 2.12.
- Remove the terminal constraint and *estimate* the domain of attraction $\hat{X}_N$ (by simulation). Compare this $\hat{X}_N$ with X_N and $\bar{X}_N$ obtained previously.
- Change the terminal cost to $V_f(x) = (3/2)x'Px$ and repeat the previous part.

Exercise 2.9: Decreasing property for the time-varying case

Consider the time-varying optimal control problem specified in 2.4.5. Suppose that $V_f(\cdot)$ and $\mathbb{X}_f$ satisfy the basic stability assumption, Assumption 2.33. Prove that the value function $V_N^0(\cdot)$ has the decreasing property

$$V_N^0((x, i)^+) \leq V_N^0(x, i) - \ell(x, i, \kappa_N(x, i))$$

for all $(x, i) \in \mathbb{X}_i$.

Exercise 2.10: Terminal cost bound for the time-varying case

Refer to Section 2.4.5. Prove that the value function $V_N^0(\cdot)$ satisfies

$$V_N^0(x, i) \leq V_f(x, i + N) \quad \forall (x, i) \in \mathbb{X}_f(i + N) \times \mathbb{I}_{\geq 0}$$

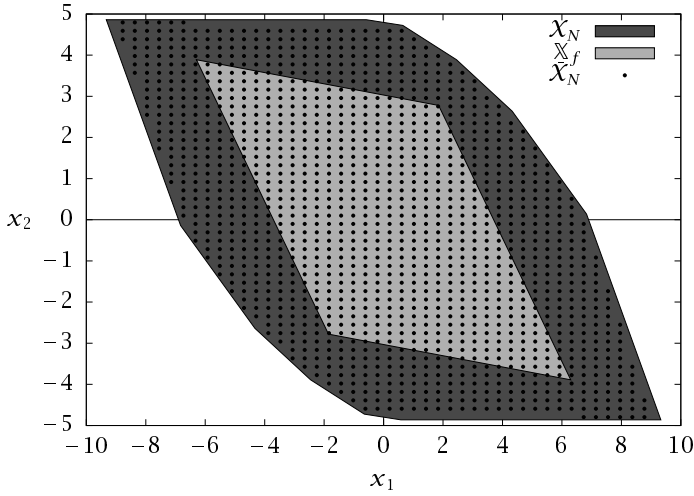


Figure 2.12: The region of attraction for terminal constraint $x(N) \in \mathbb{X}_f$ and terminal penalty $V_f(x) = (1/2)x' \Pi x$ and the estimate of $\hat{X}_N$ for Exercise 2.8.

Exercise 2.11: Modification of terminal cost

Refer to Section 2.6. Show that the pair $(\beta V_f(\cdot), \mathbb{X}_f)$ satisfies Assumption 2.14 if $(V_f(\cdot), \mathbb{X}_f)$ satisfies this assumptions, $\beta \geq 1$, and $\ell(\cdot)$ satisfies Assumption 2.2.

Exercise 2.12: A Lyapunov theorem for asymptotic stability

Prove the asymptotic stability result for Lyapunov functions.

Theorem 2.59 (Lyapunov theorem for asymptotic stability). *Given the dynamic system*

$$x^+ = f(x) \quad 0 = f(0)$$

The origin is asymptotically stable if there exist $\mathcal{K}$ functions α, β, γ , and $r > 0$ such that Lyapunov function V satisfies for $x \in r\mathcal{B}$

$$\begin{aligned} \alpha(|x|) &\leq V(x) \leq \beta(|x|) \\ V(f(x)) - V(x) &\leq -\gamma(|x|) \end{aligned}$$

Exercise 2.13: An MPC stability result

Given the following nonlinear model and objective function

$$x^+ = f(x, u), \quad 0 = f(0, 0)$$

$$x(0) = x$$

$$V_N(x, \mathbf{u}) = \sum_{k=0}^{N-1} \ell(x(k), u(k))$$

Consider the terminal constraint MPC regulator

$$\min_{\mathbf{u}} V_N(x, \mathbf{u})$$

subject to

$$x^+ = f(x, u) \quad x(0) = x \quad x(N) = 0$$

and denote the first move in the optimal control sequence as $u^0(x)$. Given the closed-loop system

$$x^+ = f(x, u^0(x))$$

- (a) Prove that the origin is asymptotically stable for the closed-loop system. State the cost function assumption and controllability assumption required so that the control problem is feasible for some set of defined initial conditions.
- (b) What assumptions about the cost function $\ell(x, u)$ are required to strengthen the controller so that the origin is exponentially stable for the closed-loop system? How does the controllability assumption change for this case?

Exercise 2.14: Stability using observability instead of IOSS

Assume that the system $x^+ = f(x, u)$, $y = h(x)$ is ℓ -observable, i.e., there exists a $\alpha \in \mathcal{K}$ and an integer $N_0 \geq 1$ such that

$$\sum_{j=0}^{N_0-1} \ell(y(j), u(j)) \geq \alpha(|x|)$$

for all x and all $\mathbf{u}$; here $x(i) := \phi(i; x, \mathbf{u})$ and $y(i) := h(x(i))$. Prove the result given in Section 2.4.4 that the origin is asymptotically stable for the closed-loop system $x^+ = f(x, \kappa_N(x))$ using the assumption that $x^+ = f(x, u)$, $y = h(x)$ is ℓ -observable rather than IOSS. Assume that $N \geq N_0$.

Exercise 2.15: Input/output-to-state stability (IOSS) and convergence

Proposition 2.60 (Convergence of state under IOSS). *Assume that the system $x^+ = f(x, u)$, $y = h(x)$ is IOSS and that $u(i) \rightarrow 0$ and $y(i) \rightarrow 0$ as $i \rightarrow \infty$. Then $x(i) = \phi(i; x, \mathbf{u}) \rightarrow 0$ as $i \rightarrow \infty$ for any initial state x .*

Prove Proposition 2.60. Hint: consider the solution at time $k + l$ using the state at time k as the initial state.

Exercise 2.16: Equality for quadratic functions

Prove the following result which is useful for analyzing the unreachable setpoint problem.

Lemma 2.61 (An equality for quadratic functions). *Let $\mathbb{X}$ be a nonempty compact subset of $\mathbb{R}^n$, and let $\ell(\cdot)$ be a strictly convex quadratic function on $\mathbb{X}$ defined by $\ell(x) := (1/2)x'Qx + q'x + c$, $Q > 0$. Consider a sequence $(x(i))_{i \in 1:p}$ with mean $\bar{x}_P := (1/P) \sum_{i=1}^P x(i)$. Then the following holds*

$$\sum_{i=1}^P \ell(x(i)) = (1/2) \sum_{i=1}^P |x(i) - \bar{x}_P|_Q^2 + P\ell(\bar{x}_P)$$

It follows from this lemma that $\ell(\bar{x}_P) \leq (1/P) \sum_{i=1}^P \ell(x(i))$, which is Jensen's inequality for the special case of a quadratic function.

Exercise 2.17: Unreachable setpoint MPC and evolution in a compact set

Prove the following lemma, which is useful for analyzing the stability of MPC with an unreachable setpoint.

Lemma 2.62 (Evolution in a compact set). *Suppose $x(0) = x$ lies in the set X_N . Then the state trajectory $(x(i))$ where, for each i , $x(i) = \phi_f(i; x)$ of the controlled system $x^+ = f(x)$ evolves in a compact set.*

Exercise 2.18: MPC and multivariable, constrained systems

Consider a two-input, two-output process with the following transfer function

$$G(s) = \begin{bmatrix} \frac{2}{10s+1} & \frac{2}{s+1} \\ \frac{1}{s+1} & -\frac{4}{s+1} \end{bmatrix}$$

- Consider a unit setpoint change in the first output. Choose a reasonable sample time, Δ . Simulate the behavior of an offset-free discrete time MPC controller with $Q = I, S = I$ and large N .
- Add the constraint $-1 \leq u(k) \leq 1$ and simulate the response.
- Add the constraint $-0.1 \leq \Delta u / \Delta \leq 0.1$ and simulate the response.
- Add significant noise to both output measurements (make the standard deviation in each output about 0.1). Retune the MPC controller to obtain good performance. Describe which controller parameters you changed and why.

Exercise 2.19: LQR versus LAR

We are now all experts on the linear quadratic regulator (LQR), which employs a linear model and quadratic performance measure. Let's consider the case of a linear model but absolute value performance measure, which we call the linear absolute regulator (LAR)⁷

$$\min_u \sum_{k=0}^{N-1} (q |x(k)| + r |u(k)|) + q(N) |x(N)|$$

For simplicity consider the following one-step controller, in which u and x are *scalars*

$$\min_{u(0)} V(x(0), u(0)) = |x(1)| + |u(0)|$$

subject to

$$x(1) = Ax(0) + Bu(0)$$

Draw a sketch of $x(1)$ versus $u(0)$ (recall $x(0)$ is a known parameter) and show the x -axis and y -axis intercepts on your plot. Now draw a sketch of $V(x(0), u(0))$ versus $u(0)$ in order to see what kind of optimization problem you are solving. You

⁷Laplace would love us for making this choice, but Gauss would not be happy.

may want to plot both terms in the objective function individually and then add them together to make your V plot. Label on your plot the places where the cost function V suffers discontinuities in slope. Where is the solution in your sketch? Does it exist for all $A, B, x(0)$? Is it unique for all $A, B, x(0)$?

The motivation for this problem is to change the quadratic program (QP) of the LQR to a linear program (LP) in the LAR, because the computational burden for LPs is often smaller than QPs. The absolute value terms can be converted into linear terms with the introduction of slack variables.

Exercise 2.20: Unreachable setpoints in constrained versus unconstrained linear systems

Consider the linear system with input constraint

$$x^+ = Ax + Bu \quad u \in \mathbb{U}$$

We examine here both unconstrained systems in which $\mathbb{U} = \mathbb{R}^m$ and constrained systems in which $\mathbb{U} \subset \mathbb{R}^m$ is a convex polyhedron. Consider the stage cost defined in terms of setpoints for state and input $x_{\text{sp}}, u_{\text{sp}}$

$$\ell(x, u) = (1/2) (|x - x_{\text{sp}}|_Q^2 + |u - u_{\text{sp}}|_R^2)$$

in which we assume for simplicity that $Q, R > 0$. For the setpoint to be unreachable in an unconstrained problem, the setpoint must be *inconsistent*, i.e., not a steady state of the system, or

$$x_{\text{sp}} \neq Ax_{\text{sp}} + Bu_{\text{sp}}$$

Consider also using the stage cost centered at the optimal steady state (x_s, u_s)

$$\ell_s(x, u) = (1/2) (|x - x_s|_Q^2 + |u - u_s|_R^2)$$

The optimal steady state satisfies

$$(x_s, u_s) = \arg \min_{x, u} \ell(x, u)$$

subject to

$$\begin{bmatrix} I - A & -B \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = 0 \quad u \in \mathbb{U}$$

Figure 2.13 depicts an inconsistent setpoint, and the optimal steady state for unconstrained and constrained systems.

- (a) For unconstrained systems, show that optimizing the cost function with terminal constraint

$$V(x, u) := \sum_{k=0}^{N-1} \ell(x(k), u(k))$$

subject to

$$x^+ = Ax + Bu \quad x(0) = x \quad x(N) = x_s$$

gives the same solution as optimizing the cost function

$$V_s(x, u) := \sum_{k=0}^{N-1} \ell_s(x(k), u(k))$$

subject to the same model constraint, initial condition, and terminal constraint.

Therefore, there is no reason to consider the unreachable setpoint problem further for an *unconstrained* linear system. Shifting the stage cost from $\ell(x, u)$ to $\ell_s(x, u)$ provides identical control behavior and is simpler to analyze.

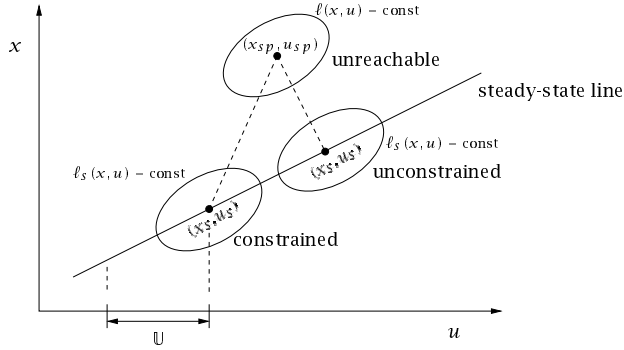


Figure 2.13: Inconsistent setpoint (x_{sp}, u_{sp}) , unreachable stage cost $\ell(x, u)$, and optimal steady states (x_s, u_s) , and stage costs $\ell_s(x, u)$ for constrained and unconstrained systems.

Hint. First define a third stage cost $l(x, u) = \ell(x, u) - \lambda'((I - A)x - Bu)$, and show, for any λ , optimizing with $l(x, u)$ as stage cost is the same as optimizing using $\ell(x, u)$ as stage cost. Then set $\lambda = \lambda_s$, the optimal Lagrange multiplier of the *steady-state* optimization problem.

- (b) For *constrained* systems, provide a simple example that shows optimizing the cost function $V(x, u)$ subject to

$$x^+ = Ax + Bu \quad x(0) = x \quad x(N) = x_s \quad u(k) \in \mathbb{U} \quad \text{for all } k \in \mathbb{I}_{0:N-1}$$

does *not* give the same solution as optimizing the cost function $V_s(x, u)$ subject to the same constraints. For *constrained* linear systems, these problems are different and optimizing the unreachable stage cost provides a new design opportunity.

Exercise 2.21: Filing for patent

An excited graduate student shows up at your office. He begins, “Look, I have discovered a great money-making scheme using MPC.” You ask him to tell you about it. “Well,” he says, “you told us in class that the optimal steady state is asymptotically stable even if you use the stage cost measuring distance from the unreachable setpoint, right?” You reply, “Yes, that’s what I said.” He continues, “OK, well look at this little sketch I drew,” and he shows you a picture like Figure 2.14. “So imagine I use the infinite horizon cost function so the open-loop and closed-loop trajectories are identical. If the best steady state is asymptotically stable, then the stage cost asymptotically approaches $\ell(x_s, u_s)$, right?” You reply, “I guess that looks right.” He then says, “OK, well if I look at the optimal cost using state x at time k and state x^+ at time $k + 1$, by the principle of optimality I get the usual cost decrease”

$$V^0(x^+) \leq V^0(x) - \ell(x, u^0(x)) \quad (2.39)$$

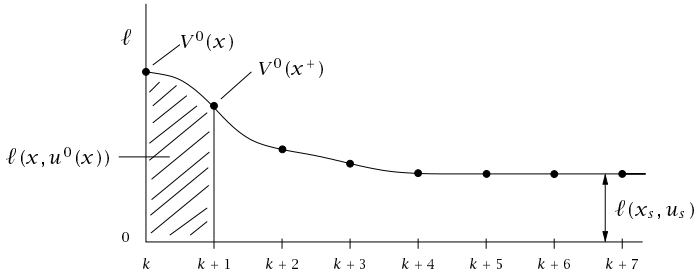


Figure 2.14: Stage cost versus time for the case of unreachable setpoint. The cost $V^0(x(k))$ is the area under the curve to the right of time k .

You interrupt, “Wait, these $V^0(\cdot)$ costs are not bounded in this case!” Unfazed, the student replies, “Yeah, I realize that, but this sketch is basically correct regardless. Say we just make the horizon *really long*; then the costs are all finite and this equation becomes closer and closer to being true as we make the horizon longer and longer.” You start to feel a little queasy at this point. The student continues, “OK, so if this inequality basically holds, $V^0(x(k))$ is decreasing with k along the closed-loop trajectory, it is bounded below for all k , it converges, and, therefore, $\ell(x(k), u^0(x(k)))$ goes to zero as k goes to ∞ .” You definitely don’t like where this is heading, and the student finishes with, “But $\ell(x, u) = 0$ implies $x = x_{sp}$ and $u = u_{sp}$, and the setpoint is *supposed* to be unreachable. But I have proven that infinite horizon MPC can reach an *unreachable* setpoint. We should patent this!”

How do you respond to this student? Here are some issues to consider.

1. Does the principle of optimality break down in the unreachable setpoint case?
2. Are the open-loop and closed-loop trajectories identical in the limit of an infinite horizon controller with an unreachable setpoint?
3. Does inequality (2.39) hold as $N \rightarrow \infty$? If so, how can you put it on solid footing? If not, why not, and with what do you replace it?
4. Do you file for patent?

Exercise 2.22: Stabilizable with small controls

Consider a time-varying system $x(i+1) = f(x, u, i)$ with stage cost $\ell(x, u, i)$ and terminal cost $V_f(x, i)$ satisfying Assumptions 2.25, 2.26, and 2.33. Suppose further that functions $f(\cdot)$ and $\ell(\cdot)$ are uniformly bounded by $\mathcal{K}_\infty$ functions α_f and α_ℓ , i.e.,

$$\begin{aligned} |f(x, u, i)| &\leq \alpha_{fx}(|x|) + \alpha_{fu}(|u|) \\ \ell(x, u, i) &\leq \alpha_{\ell x}(|x|) + \alpha_{\ell u}(|u|) \end{aligned}$$

for all $i \in \mathbb{I}_{\geq 0}$. Prove that if there exists a $\mathcal{K}_\infty$ function $\gamma(\cdot)$ such that for each $x \in \mathcal{X}_N(i)$, there exists $u \in \mathcal{U}_N(x, i)$ with $|u| \leq \gamma(|x|)$, then there exists a $\mathcal{K}_\infty$ function $\alpha(\cdot)$ such that

$$V^0(x, i) \leq \alpha(|x|)$$

for all $i \in \mathbb{I}_{\geq 0}$ and $x \in \mathcal{X}_N(i)$.

Hint: the following inequality may prove useful: for any $\mathcal{K}_\infty$ function α (see (B.1))

$$\alpha(s_1 + s_2 + \cdots + s_N) \leq \alpha(Ns_1) + \alpha(Ns_2) + \cdots + \alpha(Ns_N)$$

Exercise 2.23: Power lifting

Consider a stabilizable T -periodic linear system

$$x(i+1) = A(i)x(i) + B(i)u(i)$$

with positive definite stage cost

$$\ell(x, u, i) := \frac{1}{2} (x' Q(i)x + u' R(i)u)$$

Suppose there exist periodic control laws $K(i)$ and cost matrices $P(i)$ satisfying the periodic Riccati equation

$$P(i) = Q(i) + A(i)' P(i+1) (A(i) + B(i)K(i))$$

$$K(i) = -(B(i)' P(i+1) B(i) + R(i))^{-1} B(i)' P(i+1) A(i)$$

Show that the control law $\mathbf{K} := \text{diag}(K(0), \dots, K(T-1))$ and cost $\mathbf{P} := \text{diag}(P(0), \dots, P(T-1))$ satisfy the Riccati equation for the time-invariant lifted system

$$\begin{aligned} \mathbf{A} &:= \begin{bmatrix} 0 & 0 & \cdots & 0 & A(T-1) \\ A(0) & 0 & \cdots & 0 & 0 \\ 0 & A(1) & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & A(T-2) & 0 \end{bmatrix} \\ \mathbf{B} &:= \begin{bmatrix} 0 & 0 & \cdots & 0 & B(T-1) \\ B(0) & 0 & \cdots & 0 & 0 \\ 0 & B(1) & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & B(T-2) & 0 \end{bmatrix} \\ \mathbf{Q} &:= \text{diag}(Q(0), \dots, Q(T-1)) \\ \mathbf{R} &:= \text{diag}(R(0), \dots, R(T-1)) \end{aligned}$$

By uniqueness of solutions to the Riccati equation, this system can be used to synthesize control laws for periodic systems.

Exercise 2.24: Feasible warm start in $\mathbb{X}_f$

Establish Proposition 2.42, which states that for any $x \in \mathbb{X}_f$, the following warm start is feasible

$$\tilde{\mathbf{u}}_f(x) := (\kappa_f(x), \kappa_f(f(x, \kappa_f(x))), \dots) \in \tilde{\mathcal{U}}_N(x)$$

Recall that a warm start $\tilde{\mathbf{u}}$ is a member of $\tilde{\mathcal{U}}_N(x)$ if all elements of the sequence of controls are members of $\mathbb{U}$, the state trajectory $\phi(k; x, \tilde{\mathbf{u}})$ terminates in $\mathbb{X}_f$, and $V_N(x, \tilde{\mathbf{u}})$ is less than $V_f(x)$.

Exercise 2.25: The geometry of cost rotation

Let's examine the rotated cost function in the simplest possible setting to understand what "rotation" means in this context. Consider the discrete time dynamic model and strictly convex quadratic cost function

$$x^+ = f(x, u) \quad \ell(x, u) = (1/2)(|x - x_{\text{sp}}|_Q^2 + |u - u_{\text{sp}}|_R^2)$$

with $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, $\ell: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}_{\geq 0}$, $Q \in \mathbb{R}^{n \times n}$, $R \in \mathbb{R}^{m \times m}$ with $Q, R > 0$. We define the feasible control region as $u \in \mathbb{U}$ for some nonempty set $\mathbb{U}$. We wish to illustrate the ideas with the following simple linear system

$$f(x, u) = Ax + Bu \quad A = 1/2 \quad B = 1/2$$

subject to polyhedral constraint

$$\mathbb{U} = \{u \mid -1 \leq u \leq 1\}$$

We choose an *unreachable* setpoint that is not a steady state, and cost matrices as follows

$$(u_{\text{sp}}, x_{\text{sp}}) = (2, 3) \quad Q = R = 2$$

The optimal steady state (u_s, x_s) is given by the solution to the following optimization

$$(u_s, x_s) = \arg \min_{u, x} \{ \ell(x, u) \mid u \in \mathbb{U}, x = f(x, u) \} \quad (2.40)$$

- (a) Solve this quadratic program and show that the solution is $(x_s, u_s) = (1, 1)$. What is the Lagrange multiplier for the equality constraint?
- (b) Next we define the rotated cost function following Diehl et al. (2011)

$$\tilde{\ell}(x, u) = \ell(x, u) - \lambda'(x - f(x, u)) - \ell(x_s, u_s)$$

Plot the contour of zero rotated cost $\tilde{\ell}(x, u) = 0$ for three λ values, $\lambda = 0, -8, -12$. Compare your contours to those shown in Figure 2.15.

Notice that as you decrease λ , you rotate (and enlarge) the zero cost contour of $\ell(x, u)$ about the point (x_s, u_s) , hence the name *rotated* stage cost.

- (c) Notice that the original cost function, which corresponds to $\lambda = 0$, has negative cost values (interior of the circle) that are in the feasible region. The zero contour for $\lambda = -8$ has become tangent to the feasible region, so the cost is nonnegative in the feasible region. But for $\lambda = -12$, the zero contour has been *over rotated* so that it again has negative values in the feasible region.

How does the value $\lambda = -8$ compare to the Lagrange multiplier of the optimal steady-state problem?

- (d) Explain why MPC based on the rotated stage cost is a Lyapunov function for the closed-loop system.

Exercise 2.26: Strong duality implies dissipativity

Consider again the steady-state economic problem $\mathbb{P}_s$ for the optimal steady state (x_s, u_s)

$$\ell(x_s, u_s) := \min_{(x, u) \in \mathbb{Z}} \{ \ell(x, u) \mid x = f(x, u) \}$$

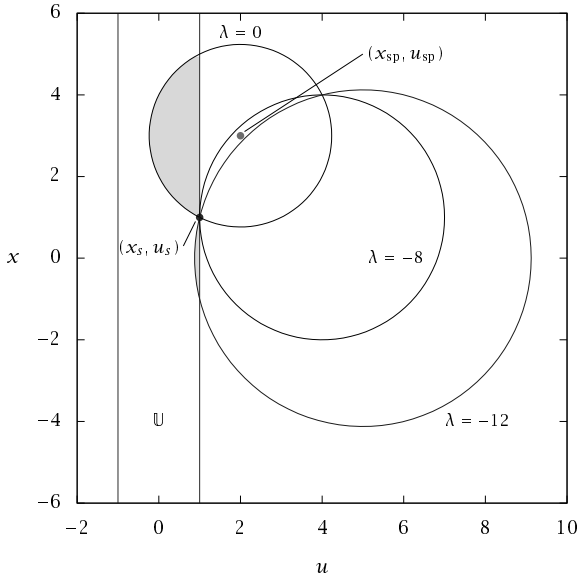


Figure 2.15: Rotated cost-function contour $\tilde{\ell}(x, u) = 0$ (circles) for $\lambda = 0, -8, -12$. Shaded region shows feasible region where $\tilde{\ell}(x, u) < 0$.

Form the Lagrangian and show that the solution is given also by

$$\ell(x_s, u_s) = \min_{(x, u) \in \mathbb{Z}} \max_{\lambda} \ell(x, u) - \lambda'(x - f(x, u))$$

Switching the order of min and max gives

$$\min_{(x, u) \in \mathbb{Z}} \max_{\lambda} \ell(x, u) - \lambda'(x - f(x, u)) \geq \max_{\lambda} \min_{(x, u) \in \mathbb{Z}} \ell(x, u) - \lambda'(x - f(x, u))$$

due to weak duality. The strong duality assumption states that equality is achieved in this inequality above, so that

$$\ell(x_s, u_s) = \max_{\lambda} \min_{(x, u) \in \mathbb{Z}} \ell(x, u) - \lambda'(x - f(x, u))$$

Let λ_s denote the optimal Lagrange multiplier in this problem. (For a brief review of these concepts, see also Exercises C.4, C.5, and C.6 in Appendix C.)

Show that the strong duality assumption implies that the system $x^+ = f(x, u)$ is dissipative with respect to the supply rate $s(x, u) = \ell(x, u) - \ell(x_s, u_s)$.

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